

TRYASIL VEGSTASJON, Norway

WMO: 013970

Lat: 61.2942N

Lon: 12.2717E

Elev: 1186

StdP: 14.08

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-8.8	-18.3	2.1	-13.1	-13.3	2.8	-8.7	14.3	31.7	11.9	31.1	1.3	130	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	79.7	59.5	75.6	59.1	71.6	57.4	63.0	73.0	60.9	70.1	59.1	68.1	5.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	79.0	65.1	57.3	73.3	63.3	55.5	68.5	61.6	29.0	72.6	27.6	69.9	26.3	67.9	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	11.8	9.9	-18.9	84.1	3.7	4.4	-21.6	87.3	-23.7	89.9	-25.8	92.4	-28.4	95.6		
(5)			-19.2	65.4	3.6	2.5	-21.8	67.2	-23.9	68.6	-26.0	70.1	-28.6	71.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	15.8	18.7	26.7	36.8	45.9	54.0	59.1	55.4	48.1	36.9	27.4	18.7		
	DBStd	16.93	11.86	10.61	9.02	5.86	7.11	5.89	5.49	4.86	5.52	6.83	8.64	12.08		
	HDD50	5413	1060	876	722	396	169	24	2	11	97	408	677	971		
	HDD65	10214	1525	1296	1187	846	593	333	197	298	506	870	1127	1436		
	CDD50	691	0	0	0	1	41	144	283	179	40	3	0	0		
	CDD65	17	0	0	0	0	0	2	14	1	0	0	0	0		
	CDH74	494	0	0	0	0	31	146	285	32	0	0	0	0		
	CDH80	80	0	0	0	0	0	23	55	2	0	0	0	0		
	WSAvg	4.2	3.3	3.6	4.4	4.6	5.0	5.3	4.6	4.1	4.2	3.9	3.7	3.4		
	PrecAvg	32.6	2.2	1.6	1.3	1.8	3.0	3.3	3.7	4.3	3.1	3.3	2.9	2.4		
	PrecMax	44.3	5.0	3.9	2.8	4.2	6.4	5.7	6.9	5.9	5.7	7.3	8.4	5.4		
(15) Precipitation	PrecMin	26.9	0.4	0.4	0.2	0.4	0.9	1.9	1.0	1.7	1.1	0.6	1.1	0.5		
	PrecStd	4.2	1.1	0.7	0.7	1.0	1.3	1.1	1.6	1.1	1.4	1.7	1.6	1.1		
	DB	38.9	42.4	53.7	64.5	78.3	83.2	85.0	78.1	69.5	59.0	48.4	42.7			
	MCWB	36.5	37.0	43.1	49.4	56.9	57.8	63.4	60.4	57.0	52.8	46.2	39.8			
	2% DB	36.1	37.5	47.0	59.8	72.3	78.7	81.0	73.1	65.2	53.5	44.6	39.4			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	33.9	34.6	38.9	46.3	54.7	56.9	61.9	58.9	54.8	48.6	42.8	37.2			
	5% DB	33.4	34.7	42.6	54.7	67.6	73.2	77.0	69.7	61.9	50.5	41.4	36.6			
	MCWB	31.9	32.7	35.9	43.4	52.8	56.7	60.3	57.5	53.4	46.6	40.1	34.7			
	10% DB	31.9	32.7	38.9	49.5	61.3	68.3	73.0	66.2	58.5	47.7	38.1	33.9			
	MCWB	30.9	31.5	34.2	40.4	49.6	54.2	58.9	56.3	52.0	44.7	36.7	32.7			
	0.4% WB	36.8	38.2	43.5	50.3	59.7	62.7	66.4	63.6	59.1	54.1	47.1	40.5			
	MCDB	38.3	40.8	52.3	62.2	70.7	72.1	76.5	70.4	64.9	57.3	48.4	42.1			
	2% WB	34.5	35.1	39.4	47.1	56.9	59.6	64.1	61.5	56.7	49.9	43.2	37.7			
(30) Mean Coincident Dry Bulb Temperatures	MCDB	35.8	37.1	45.4	57.7	69.7	71.1	76.6	68.1	61.9	52.9	44.6	39.2			
	5% WB	32.4	33.1	36.8	44.0	54.2	57.7	62.2	59.6	54.8	47.3	40.4	34.9			
	MCDB	33.2	34.3	41.2	54.0	64.5	70.1	73.3	66.5	60.1	49.7	41.4	36.1			
	10% WB	30.8	31.6	34.7	41.2	51.0	55.7	60.3	57.8	52.9	45.0	37.0	32.7			
	MCDB	31.9	32.8	38.5	48.5	60.1	66.4	70.3	64.0	57.5	47.4	38.0	33.7			
	MDBR	13.1	16.0	20.2	20.9	21.5	21.9	21.1	19.4	18.9	14.1	10.1	12.6			
	5% DB	11.3	15.9	23.7	31.2	34.0	32.8	29.7	26.2	25.0	18.5	12.0	12.5			
	MCWBR	10.3	13.0	16.1	18.9	17.7	15.2	13.3	14.0	15.6	13.6	10.6	11.3			
(38) Clear-Sky Solar Irradiance	5% WB	10.2	13.7	21.3	29.7	27.7	26.6	25.2	20.4	21.2	15.1	10.9	12.1			
	MCWBR	9.6	11.4	14.7	18.4	15.3	13.4	12.1	11.9	14.2	12.0	10.0	11.2			
	taub	0.222	0.271	0.301	0.339	0.346	0.339	0.358	0.348	0.317	0.301	0.241	N/A			
	taud	2.134	2.249	2.253	2.303	2.434	2.482	2.441	2.460	2.534	2.479	2.225	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	167	228	259	271	279	282	273	266	257	219	157	N/A			
	Edh at Noon	11	20	29	33	32	31	31	28	22	16	10	N/A			
(44) All-Sky Solar Radiation	RadAvg	90	300	794	1296	1492	1662	1503	1170	763	342	107	49			
	RadStd	13	59	78	132	144	138	199	108	93	60	27	5			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page